

Audio file

[6.2 Acute Leukemia.mp4](#)

Transcript

Welcome back. The next discussion that we're going to have concerns acute leukemia. However, before we can understand acute leukemia, we need to briefly review hematopoiesis. So let's remind ourselves that hematopoiesis begins with a hematopoietic stem cell, and that hematopoietic stem cell eventually develops into a myeloid blast, or a lymphoid blast. The myeloid blast is also known as the myeloid stem cell. The lymphoid blast is also known as the lymphoid stem cell. And the myeloid blast eventually has the ability to produce multiple myeloid lineages. For example, it can develop into a erythroblast that would produce the red blood cells, and it can develop into a monoblast that would produce the monocytes. It can develop into a myeloblast which would produce the granulocytes, and it can develop into a megakaryoblast that would produce the megakaryocytes. Similarly, the lymphoblast has the ability to eventually develop into a B lymphoblast, which would produce B cells, and a T lymphoblast, which would eventually produce T cells. Now, this stepwise maturation of the hematopoietic stem cell to a myeloid blast and eventually to red blood cells, et cetera, is constantly occurring. In acute leukemia, the idea is that there is a disruption in the ability for the cells to mature, so that the myeloblast, for example, or the lymphoblast, for example, loses the ability to become a more mature cell, and when it loses the ability to become a more mature cell, this cell begins to pile up. And that's the hallmark of acute leukemia. As I've stated here in your text, acute leukemia is a neoplastic proliferation of blasts. Now, we should remember that if I were to take my bone marrow or your bone marrow and to look for blasts, I would see a certain percentage of blasts. In fact, 1 to 2% is the normal number. And the reason we see blasts within our bone marrow is because our bone marrow has blasts, and those blasts are the cells that are generating the more mature cells that will eventually be pushed off into the blood. So we need a strict definition for what exactly an acute leukemia is, and that definition is greater than 20% blasts in the bone marrow. When there are greater than 20% blasts in the bone marrow, that is defined as an acute leukemia. Now, in an acute leukemia, the blasts will crowd out normal hematopoiesis, and so the patients will classically present with signs and symptoms of loss of the cells that are normally produced by the bone marrow. For example, if there's a loss in red blood cells, the patient will develop anemia. If platelets can't be formed, the patient will develop thrombocytopenia. And if the neutrophils can't be formed, the patient will develop neutropenia. And so, of

course, if they have anemia, they would present with signs and symptoms of hypoxia. If they have thrombocytopenia, they would present with signs and symptoms of bleeding. And if they have neutropenia, they would present with signs and symptoms of infection. But the notion here is that as these blasts begin to pile up within the bone marrow, they crowd out the normal hematopoiesis, resulting in the classic signs and symptoms. These signs and symptoms tend to be acute. The patients present all of a sudden with these signs and symptoms, and hence the term acute leukemia. At the same time, the blasts usually enter into the blood, and when they enter into the blood, they result in a high white blood cell count. If we were to look at these blasts on a blood smear or even on a bone marrow aspirate, we would see that the blasts are large, immature cells with a punched-out nucleolus. I've included in your text a picture of a blast. A few important things. First of all, these are large cells. Here's the normal red blood cell. Clearly, these cells are much larger than the normal red blood cell. So one feature of a blast is that it is a large cell. It's relatively immature. It doesn't have much cytoplasm. And the third important feature, as I've already mentioned, is that there is a nucleolus. If you look at the nucleus of this cell, it appears as if someone took a hole puncher and punched out a little circular region of that nucleus. That represents the nucleolus, that punched out area. Now, when we have an acute leukemia, if I were to look at this blood smear, I could clearly identify these as blasts. However, I may not be able to tell you whether these are lymphoblasts or whether these are myeloblasts. And so that requires a little bit of explanation. Again, remember that the hematopoietic stem cell is going to divide into myeloid blast and a lymphoid blast. Now, if there's an accumulation of the myeloid blast, that is called AML. And if there's an accumulation of the lymphoid blast, that is called ALL. When we look at a blast on a bone marrow aspirate or on a blood smear, it's important for us to be able to determine whether that cell represents a myeloblast or whether that cell represents a lymphoblast, because these are two very different diseases with very different treatments. By looking at the cell, it's very difficult to tell whether we're dealing with a myeloblast or a lymphoblast. So instead, we use certain markers. The key marker, the hallmark marker for a lymphoblast is going to be positivity for TDT in the nucleus. TDT is a DNA polymerase and it is present in the nucleus of lymphoblasts and only in the nucleus of lymphoblasts, meaning it's not present in mature lymphocytes and it's not present in the myeloid blasts. And so TDT is the key marker that's going to identify for us that we're dealing with a lymphoblast. At the same time, one of the key markers for a myeloid blast, sort of the classic key marker for a myeloid blast, is the presence of myeloperoxidase. Now, myeloperoxidase can be detected in two ways. The first way would be to do a chemical study to determine that this enzyme is present in the cytoplasm of the cells. And the second way would be to actually view the myeloperoxidase under the microscope. We can't see myeloperoxidase under the microscope. However, myeloperoxidase can sometimes crystallize into a special structure called an Auer rod.

And so the presence of an Auer rod in a cell indicates that we basically are dealing with AML. Now, that's what I say here in your slides, that acute leukemia is subdivided into ALL and AML, and that's based on the phenotype of the blasts. ALL is a neoplastic accumulation of lymphoblasts, as I've already explained, and the key finding for a lymphoblast is going to be positivity for TDT. Again, when you have blasts and you want to decide whether it's a lymphoblast or a myeloid blast, TDT is a key marker because TDT indicates that you're dealing with a lymphoblast. It's important to know for examination purposes that TDT is a DNA polymerase, and obviously that DNA polymerase would be present within the nucleus. TDT, as I've already stated, is not present in myeloid blasts and is not present in mature lymphocytes, and so it's a very helpful marker. Of course, ALL most commonly arises in children. One very high-yield association is that ALL can be associated with Down syndrome, but after the age of five, and that's going to be important a little later in our discussion. Patients who have Down syndrome can develop ALL, but usually it's after the age of five. Now, ALL becomes a little more complicated in that it can be subclassified into BALL and TALL. Remember that the hematopoietic stem cell if we look at the lymphoid lineage, will become a lymphoblast, and that lymphoblast will become a B lymphoblast or a T lymphoblast. And of course, the B lymphoblast will produce B cells, and the T lymphoblast will produce T cells. So what we're saying here is that when you have ALL, you can actually have a B ALL or a T ALL. And this is important because it also has some clinical ramifications. So as I say here, ALL can be subclassified into BALL and TALL, and we base that on surface markers. The way by which we make that distinction is surface markers. In other words, both BALL and TALL will have TDT positivity. That allows it to be called an ALL, but the surface markers will distinguish the two. So BALL is the most common type of ALL, and the key classical surface markers The classical surface markers are CD10, CD19, and CD20. These are the classic markers seen in a BALL. BALL has an excellent response to chemotherapy, but it's important to remember that it requires prophylaxis to the scrotum and the CSF. When we give chemotherapy to these patients, the chemotherapy obviously goes into the blood, but it's not able to properly cross the blood-brain barrier or the blood-testicle barrier. These are special sites that require chemotherapy to be directly injected within them. And so, prophylactically, the patients will get chemotherapy to the scrotum and to the CSF. Here's a picture of the meninges of the brain. This represents the meninges, and all of this blue, all these blue dots within the meninges here. This all represents ALL. Again, I just put this picture here to remind you that patients who have ALL require prophylactic chemotherapy to the scrotum and to the CSF. The prognosis for BALL is generally good, and it's based on cytogenetic abnormalities. The classic BALL that we're all familiar with, that usually has cytogenetic abnormality of translocation between chromosomes 12 and 21, 12, 21, sort of 12 flipped over is 21. and this has a good prognosis. And this is the one that we more commonly see in kids. Now, there is another

subtype of ALL in which patients can actually have a 9/22 translocation, and the 9/22 translocation has a poor prognosis. Now, remember that the 9/22 translocation is classic and is a defining feature of CML. However, it can also be seen in a subset of patients that have BALL, and this portends a poor prognosis. This is also known as pH-positive ALL. pH stands for the Philadelphia chromosome, so remember that the 922 translocation is also called the Philadelphia chromosome, and this subtype is more commonly seen in adults. tALL would be categorized by lymphoblasts, which are going to be TDT positive, that express a marker that ranges from the number CD2 up to CD8. For example, CD3, CD4, CD8, CD5, CD7, these are all markers of T-cell ALL. Important to note that the blasts do not express CD10. Now, T-ALL T-A-L-L is very easy to remember because the T is sort of the mnemonic. So it presents as a thymic mass, and that makes sense because that's where the T-cells normally would go. So it's going to be in the mediastinum, and it presents in a teenager. So thymic mass and teenager. At the same time, because it's a thymic mass instead of cells floating around in the blood, we call this acute lymphoblastic lymphoma instead of leukemia. Recall that leukemia means that the malignant cells would be floating around in the blood, and lymphoma, oma means mass, lymphoma means that the malignant cells are forming a mass. So we would call this disease acute lymphoblastic lymphoma because it would be classically forming a mass in the mediastinum. Moving on to AML, AML is a neoplastic accumulation of myeloblasts. And the myeloblasts, as I've already stated, would be characterized by myeloperoxidase. Myeloperoxidase can be detected in two ways. One way would be to chemically detect the myeloperoxidase using special stains, and the second way would be to visually see the myeloperoxidase as crystal aggregates called Auer rods. And so the presence of Auer rods within a blastic cell indicates that we're dealing with AML. Let's look at a picture. Here is a classic picture of a blast. This is a large cell, much bigger than the adjacent red blood cells, and you can notice here that there's a punched-out nucleolus. Along with this punched-out nucleolus, there is this structure present in the cytoplasm, which is an auer rod, and this auer rod indicates that the cell must be a myeloblast because an auer rod is basically an aggregate of MPO. Now, AML is most commonly seen in older adults. The average age is about 55 years. And AML can be subclassified in three ways. One way would be to look at cytogenetic abnormalities. And this is the predominant mechanism by which we subclassify AML. So the first goal in an AML is to try to determine if there is a recurring cytogenetic abnormality. If we can't find a classic cytogenetic abnormality, the lineage of the myeloblast can also be used to subclassify AML. And finally, surface markers are often important in classifying AML as well. A classic type of AML is that AML which is characterized by a 15-17 translocation. And this is also known as acute promyelocytic leukemia. very high yield to know that the 15-17 translocation results in disruption of the retinoic acid receptor. When this retinoic acid receptor is disrupted, this then disrupts the ability of the cells to mature.

And so instead of maturing, promyelocytes will accumulate. The promyelocytes contain numerous auer rods, and that's important because the auer rods have the potential of activating the coagulation cascade, and so there is a risk for DIC. Acute promyelocytic leukemia is a medical emergency, and that's because of the risk of DIC. The treatment for acute promyelocytic leukemia is also very important to be aware of. If the patient is given ATRA, which is all trans-retinoic acid, it's a derivative of vitamin A, this will then bind the disrupted retinoic acid receptor and caused the blast to mature and to become neutrophils. And of course, that would alleviate the leukemic burden in that particular patient. So these are all very high yield principles. 15/17 translocation disrupts the retinoic acid receptor, causing promyelocytes to accumulate. The promyelocytes contain hour rods. The hour rods increase the risk for DIC. If ATRA is given, ATRA will then bind the retinoic acid receptor, causing the cells to mature and alleviating the patient of their leukemic burden. Now, recall that I said that if there is no recurring cytogenetic abnormality, then we subclassify the AML based on the lineage of differentiation. That requires us to review the lineage of differentiation briefly. Remember that we begin with a hematopoietic stem cell. The hematopoietic stem cell eventually becomes a myeloid stem cell, also called a myeloid blast. Now, this myeloid blast has the ability to produce an erythroblast, or a myeloblast, or a monoblast, or a megakaryoblast. And of course, each of these cells would then produce their relevant, mature offspring. Now, when we globally think about AML, we could get an AML that actually is composed of erythroblast. That's called an erythroblastic AML. We could get a monoblastic AML. We could get a megakaryoblastic AML, et cetera. And so when we don't have a recurring cytogenetic abnormality, we will subclassify the AML based on the lineage of differentiation of the cells. This actually results in a slightly complicated classification of leukemia. However, there are a few high-yield subtypes to be familiar with, and we're only going to touch on those. So one high-yield subtype to be familiar with is acute monocytic leukemia. What's characteristic of acute monocytic leukemia is that you get a proliferation of monoblasts. And what's very important and super high yield in this discussion is that these monoblasts, they characteristically infiltrate the gums so that the patients will present with involvement of the gums. And that's what this picture here is trying to highlight. This is a patient, and they've got an acute monocytic or monoblastic leukemia, and clearly there is swelling of the gums. Another important high-yield subtype to be aware of is acute megakaryoblastic leukemia. This would be a proliferation of megakaryoblasts. Megakaryoblasts do not have MPO. Obviously, megakaryocytes don't need MPO because they're not going to be doing the oxygen-dependent killing that MPO is useful for. And so megakaryoblasts would lack MPO, but what's even more important than that is that there's an association with Down syndrome, but before the age of five. So this then brings to a general principle. Patients with Down syndrome have an increased risk of acute leukemia. The acute leukemia is more

commonly acute megakaryoblastic leukemia before the age of five and is more commonly ALL after the age of five. So we should put that association together based on what we've learned through this lecture. It is also possible to get an AML from pre-existing dysplasia, and the classic circumstance in which there would be pre-existing dysplasia would be in a patient that has a history of exposure to alkylating agents, for example, chemotherapy or radiotherapy. Now, what happens in these patients is that they tend to develop something called a myelodysplastic syndrome. In a myelodysplastic syndrome, the patients get a hypercellular bone marrow, but because the cells are not being formed properly, dysplastic, bad growth, they actually don't get out into the blood, and so the patients have cytopenias, low counts of cells in the blood. In this particular case, if you were to do a bone marrow biopsy, you would see increased blasts. However, those increased blasts would be less than 20%. Of course, greater than 20%, we would not call it myelodysplastic syndrome, we would call it acute leukemia. Most of these patients, because they have cytopenias, will actually die from infection or bleeding, but a subset of them can progress to acute leukemia, such that if the blasts become greater than 20%, then we would say that the patient has, for example, AML that has arisen in a background of myelodysplasia. So that concludes our discussion of acute leukemia, and the next discussion will then be chronic leukemia.